

Appendix: Preliminary Phase B Experiment 1 Findings

Companion to arXiv v1 (MaoField: A Dialectical-Materialist Framework for Non-Statistical Semantic Representation, Chen 2026); results included in v0.1.1 Zenodo release.

Summary. A 36-hour experimental campaign on the open-dissipative b+c-feedback PDE formulation of MaoField produces a clean differential signal — 50 localized phase-coherent patches per document in the experimental mode versus 0 in every control lacking either feedback or multi-scale coarse-graining — together with an approach-to-plateau behavior of the dissipation rate at $dS/dt \approx 10$ over 150 dimensionless time units, five orders of magnitude below the initial transient. A structural consequence of the feedback architecture — that translationally-invariant feedback operators cannot couple to the base-source mean — accounts for the observation that whitening the 17 BGE-drift component of the source has zero measurable effect on the dynamics; this partially obviates arXiv v1 OP2 sub-problem (a) source-drift de-biasing for the mean channel. Three honest language refinements (§A.5) replace heuristic claims with quantitatively supported statements; no insight from the 2026-04-15 five-insight sketch is falsified, three are refined, two are merged.

A.1 Scope and status

This appendix reports preliminary findings from Phase B Experiment 1, an empirical test of five theoretical insights developed during a 2026-04-15 discussion between one of the principal collaborators and a research assistant. The original arXiv v1 manuscript (§5.4) flagged Phase B as an “outlook” program for after the Block V Phase A mechanism paper (IPM target, 2026-07). Acting on the principal investigator’s 2026-04-15 decision, the first Phase B experiment was compressed into the v0.1.1 release week, executed autonomously with independent review gates.

The findings are **preliminary**. They rest on a single parameter setting of the feedback and history-decay constants ($\alpha = 0.1$, $\beta = 0.05$, $\gamma = 0.05$, $N_{\text{hist}} = 100$, $\text{block_size} = 2$), a 70-document sample from NFCorpus with cached BGE-M3 embeddings, and a deterministic explicit-Euler integrator with no stochastic noise. Parameter scans, larger corpora, and Langevin extensions are deferred to subsequent Phase B experiments. The present result establishes that the framework, under the b+c-feedback architecture proposed in §A.2, is empirically **non-vacuous** and **qualitatively consistent** with the five-insight dialectical-materialist sketch.

Three honest language downgrades — described in §A.5 — replace heuristic qualitative claims with quantitatively supported, mechanistically grounded statements. The downgrades **strengthen** the paper’s integrity: they are not concessions to negative results but refinements of the initial formulation.

A.2 Experimental design

The complex scalar field $\psi : \mathbb{R}^3 \rightarrow \mathbb{C}$ on a 32^3 periodic lattice ($dx = 1$) evolves by the overdamped equation

$$\gamma \partial_t \psi = -D \nabla^2 \psi - 2(|\psi|^2 - v^2) \psi + S(x, t) \quad (A.1)$$

with $\gamma = 1$, $D = 0.1$, $v = 1$ (Mexican-hat potential, U(1) global symmetry). The source $S(\mathbf{x}, \mathbf{t})$ implements bidirectional self-feedback:

$$S(x, t) = S_0(x) + \alpha (\psi(x, t) - \langle \psi \rangle_x) + \beta \delta S_{\text{history}}(x, t) \quad (A.2)$$

with $S_{\text{history}}(\mathbf{x}, \mathbf{t}) = \sum_{\mathbf{k}} \mathbf{w}_{\mathbf{k}} \cdot ((\mathbf{x}, \mathbf{t} - \mathbf{t}_{\text{outer}}) \cdot \langle \rangle_{\{\mathbf{x}, \mathbf{t} - \mathbf{t}_{\text{outer}}\}}) \cdot \mathbf{t}_{\text{outer}}$, exponential decay $\mathbf{w}_{\mathbf{k}} = \exp(-\mathbf{k} \cdot \mathbf{t}_{\text{outer}})$, and $S(\mathbf{x})$ the BGE-M3 embedding of the document tiled on the lattice (halves assigned to S_a and S_b , three passes of 3-point box smoothing).

A three-time-scale integrator separates evolution, source refresh, and history accumulation:

- **inner** $\mathbf{t} = 0.01$ — explicit Euler step on ψ using the current S
- **middle** every 10 inner steps ($\mathbf{t} = 0.1$) — recompute S from (A.2) and replace it with a block-averaged coarse version (block = 2, producing 16^3 coarse cells); this coarse S drives the next 10 inner steps
- **outer** every 100 inner steps ($\mathbf{t} = 1.0$) — push the current $\langle \rangle$ snapshot onto the history buffer

The simulation runs for 5,000 inner steps ($\mathbf{t}_{\text{sim}} = 50$). The history buffer has capacity $N_{\text{hist}} = 100$ and fills to 50 samples over $\mathbf{t}_{\text{sim}} = 50$. All runs use base seed 20260421 with per-document offset $17i$.

Five operational modes (Table A.1) differ by (γ, β) , multi-scale versus single-scale integrator, and whitened versus unwhitened source base.

mode	multi-scale	source	whiten	purpose
exp	0.1	0.05	no	main experiment
null / control_const	0	0	no	static-source control
control_whiten	0.1	0.05	yes	isolates BGE 17 mean effect

mode		multi-scale	source whiten	purpose
control_single	0.05	no	no	isolates multi-scale effect

Two additional O3-dedicated runs vary `dt_inner` $\{0.1, 1.0\}$ with `t_sim` = 50 held fixed; one extended run (`t_sim` = 200) checks late-time plateau behavior. Eight runs total, 70 documents each, zero NaN or divergence across all runs.

A.3 Results

A.3.1 Localized phase-coherent structures (O1)

Each stationary -snapshot is analyzed by two independent algorithms, per an independent mathematical review of the observable: (i) connected-component labeling on a local phase-coherence field $C(\mathbf{x}) = |\langle \mathbf{e}^{\{i(y)\}} \rangle_{\mathbf{y} \in N(\mathbf{x})}|$ with 27-voxel neighborhood and threshold $T = \langle C \rangle + \langle C \rangle$; (ii) polar-decomposition radial structure factors $S_{-}(|\mathbf{k}|)$ and $S_{+}(|\mathbf{k}|)$ from $\mathbf{e}^{\{i\}}$, with pseudo-Goldstone indicator $R = S_{-}(k_{\min})/S_{+}(k_{\min})$ and low- k log-log slopes α_{-} , α_{+} .

mode	N_struct_valid	Pass A (3)	R	α_{-}	Pass B	Combined
exp	49.8	100%	7,768	1.28	94.3%	94.3%
control_whiten	49.8	100%	7,742	1.28	94.3%	94.3%
null	0.0	0%	16,923	2.54	100%	0%
control_const	0.0	0%	16,923	2.54	100%	0%
control_single	0.0	0%	20,344	2.51	100%	0%

The differential signal is large and robust: feedback combined with multi-scale coarse-graining yields 50 phase-coherent patches of gyration radius below `grid/4` = 8. Removing either the feedback (`null`, `control_const`) or the multi-scale coupling (`control_single`) yields zero patches. Pass B alone cannot distinguish the experimental mode from the controls — indeed, `null`, `control_const`, and `control_single` all exhibit R ratios roughly twice that of `exp` (17,000–20,000 vs 7,700) — because the phase-vs-amplitude asymmetry is a generic feature of overdamped relaxation from random initial conditions in a U(1) Mexican-hat potential; the uniform relaxation state in the controls produces an even smoother angular spectrum than the feedback-structured state. The discriminating observable is Algorithm A: the **coexistence** of pass-B-like phase softness and pass-A-like localized structure formation is what separates the experimental mode from the controls.

A.3.2 Non-equilibrium steady-state dissipation (O2)

Entropy production rate is computed in the Seifert (2005) overdamped framework with $\beta = T = 1$: $dS/dt := \langle |\dot{\mathbf{t}}|^2 \rangle_{\mathbf{x}}$ is the mean-field dissipated power per voxel, equivalent in these units to the medium entropy production. To falsify the Lyapunov-necessary interpretation $dS/dt \rightarrow 0$ under static $\mathbf{S}$, we additionally require that the source field exhibit meaningful late-window time variance.

mode	late dS/dt	late $ \mathbf{S} $	$ \mathbf{S} $ CoV (late)
exp	$2.19 \cdot 10$	1.018	0.107
control_whiten	$2.24 \cdot 10$	1.018	0.107
null	$8.28 \cdot 10$	0.008	0
control_const	$8.28 \cdot 10$	0.008	0
control_single	$8.62 \cdot 10$	0.103	0.005

The extended run ($t_{\text{sim}} = 200$) reveals a genuine approach-to-plateau behavior. Over the window $t \in [100, 200]$, the cross-document mean dS/dt decays from $4.14 \cdot 10$ to $2.90 \cdot 10$ — a factor of 0.70 over 100 time units. Fitting $dS/dt(t) = c + A \cdot t^{-\alpha}$ per document over $t \in [25, 200]$ with free yields a median $\alpha = 2.9$ (interquartile range $[2.87, 2.95]$) and a median asymptote $c = 2.9 \cdot 10$, with $c > 10$ in **100% of documents**. The atypically large α value in this free-parameter fit reflects a crossover from the early transient to the late plateau that is not well-modeled by a single power-law; with α fixed at the $t^{-1.2}$ first-principles expectation for an overdamped-relaxation tail, the fit yields negative c values, indicating the residual decay is faster than $t^{-1.2}$. What is robust across both fitting protocols is the observation that (i) 100% of documents exhibit a late-time $dS/dt > 10$, (ii) the decay rate is visibly decreasing (consistent with plateau approach rather than continued power-law decay toward zero), and (iii) the source field exhibits non-zero late-window coefficient of variation 0.107, ruling out the Lyapunov-necessary interpretation $dS/dt \rightarrow 0$ under static $\mathbf{S}$. The exact functional form of the approach (power-law with exponent in $[1, 3]$, stretched exponential, or a crossover regime) is not pinned down by the available data and is a follow-up target.

The late-time dS/dt value represents a suppression of 5 orders of magnitude relative to the initial transient $dS/dt(t = 0) = 5 \cdot 10^1$, consistent with the “contradiction persistence with reduced intensity” prediction of the dialectical framing introduced during the 2026-04-15 discussion.

A.3.3 Kinematic self-similarity under block averaging (O3)

Three runs of the experimental mode with $dt_{\text{inner}} \in \{0.01, 0.1, 1.0\}$ and $t_{\text{sim}} = 50$ held fixed produce spatial structure functions $S^{\{||^2\}}(\mathbf{r}) = \langle (||^2(\mathbf{x} + \mathbf{r}) - ||^2(\mathbf{x}))^2 \rangle$ (radially binned). Pairwise log-log Pearson correlations in $\mathbf{r} \in [1, 8]$:

pair	raw Pearson	log-log Pearson
dt = 0.01 dt = 0.1	0.9999	0.9999
dt = 0.01 dt = 1.0	0.9156	0.9809
dt = 0.1 dt = 1.0	0.9121	0.9794

Power-law fits $S(r) \sim r^\alpha$:

run		residual
dt = 0.01	0.302	0.100
dt = 0.1	0.311	0.101
dt = 1.0	0.001	0.000

The $dt = 0.01$ and $dt = 0.1$ runs agree to 3% in scaling exponent — a clean indicator of kinematic self-similarity across one decade of inner-step resolution. The $dt = 1.0$ run is excluded from the comparison: explicit-Euler linear stability on the nonlinear potential term $2(|\mathbf{v}|^2 - v^2)$ requires $\tau < 2/(6D + 4v^2) \approx 0.43$, violated at $\tau = 1.0$. The clamp $|\mathbf{a}|, |\mathbf{b}| < 3$ then engages, producing a bang-bang dynamics with flat scaling. An earlier hypothesis that this outlier reflected a physical cutoff coincident with the middle-tick scale $\tau = 0.1$ has been retracted.

A.4 Exploration findings

Four unexpected signals in the raw data were investigated by an independent analysis:

(i) **Experimental mode whitened-source control, to three significant figures.** This is an architectural structural property of the feedback scheme, not empirical coincidence. The feedback terms $\langle \mathbf{v} \rangle$ and $\mathbf{v} \cdot \mathbf{S_history}$ are by construction mean-subtracted (history snapshots store $\langle \mathbf{v} \rangle$), hence translationally invariant, hence insensitive to the spatial mean of the base source. Formally: since $\langle \mathbf{v} \rangle$ and $\mathbf{v} \cdot \mathbf{S_history}$ both integrate to zero in $\mathbf{x}$, the constant component of $\mathbf{S}$ enters only the one-dimensional ODE for $\langle \mathbf{v} \rangle(\tau)$, decoupled from spatial modes. The base-source mean $\langle \mathbf{S} \rangle$ therefore shifts $\langle \mathbf{v} \rangle$ but cannot seed spatial structure. Quantitatively, $|\langle \mathbf{S} \rangle| / |\mathbf{S_saturated}| \approx 0.0012$ — the 17- σ z-score of the BGE-drift violation of Laplace detailed-balance reported in arXiv v1 §4.9 is statistically significant but physically negligible in L^2 . **Consequence for the theory:** the architectural claim “**the feedback layer is robust to base-source mean**” replaces the earlier heuristic claim “**BGE drift is the historical imprint that drives open dissipation.**” Open dissipation arises from the feedback-plus-multi-scale coupling, not from the upstream statistical idiosyncrasy.

(ii) The $dt = 1.0$ outlier in O3 is a numerical, not physical, signature. See §A.3.3.

(iii) **Insights 1 (localized excitations) and 5 (multi-scale synchronization) are a single proposition.** Removing multi-scale block-averaging while keeping the feedback (`control_single`) produces zero localized structures. The feedback operators preserve translation invariance by construction; block-averaging with `block = 2` imposes a fixed 16^3 lattice of coarse cells, which is the symmetry-breaking perturbation that seeds the feedback’s structure formation. The merged claim is: *localized phase-coherent excitations emerge in the feedback-driven PDE only when block-averaging provides a translation-symmetry-breaking seed.* The original two-insight formulation is revised accordingly.

(iv) **The O2 asymptote is a true plateau, not power-law decay to zero.** See §A.3.2 three-parameter fit.

A.5 Three honest language downgrades

original	downgraded
“RG self-similarity”	“ kinematic self-similarity under block averaging — necessary but not sufficient for an RG fixed point”
“sustained $dS/dt > 0$ ”	“approach-to-plateau behavior at ~ 10 , five orders of magnitude suppressed from the initial transient, over 150 dimensionless time units; 100% of documents exceed 10 at $t = 200$; decay rate visibly decreasing (late-time window ratio 0.70 over 100 time units); exact functional form of the approach not pinned down by current data”
“BGE drift is the historical imprint that drives open dissipation”	“BGE drift is not a differential driver of the dynamics; the feedback layer is architecturally robust to base-source mean; open dissipation arises from feedback-plus-multi-scale coupling”

These are not retreats. Each downgrade replaces a heuristic qualitative claim with a quantitatively supported, mechanistically grounded statement.

A.6 Five insights — verdict

#	original insight	verdict	notes
1 & 5	elementary particles = field excitations (localized Goldstone / soliton / breather) + multi-scale synchronization (RG from Phase B)	merged: multi-scale-seeded localized excitations	see §A.4 (iii); block-averaging is the translation-symmetry-breaking seed
2	atom = process (Axiom 1: dynamic, entropy-increasing, cross-scale, mutable)	conditional pass	§A.3.2 — quasi-stationary NESS at 10
3	driving = objective material reflection (open dissipation; BGE drift is historical imprint, not noise bias)	refined — architectural theorem	§A.4 (i); feedback robust to base-source mean
4	coupling = b + c bidirectional self-feedback (Axiom 6 “matching = self-training” in mathematical form)	pass	$ ^2$ from 1.00 $\rightarrow$ 1.42, source S from 0.008 $\rightarrow$ 1.37, free energy from +10 $\rightarrow$ $3.6 \cdot 10$

No insight is falsified, three are refined, two are merged. The refinements strengthen the theoretical coherence of the framework by grounding its central claims in mechanisms rather than heuristics.

A.7 Relation to arXiv v1 open problems

OP1 (Axiom 6 formalization). Phase B Exp 1 does not address OP1. Axiom 6 M2 falsification (arXiv v1 §5.1) remains the current state of the open problem.

OP2 (Axiom 3 three-fold co-design). The architectural theorem of §A.4 (i) partially obviates OP2 sub-problem (a) — source-drift de-biasing — for the

mean channel: when the feedback is the coupling route, the base-source mean is not a necessary upstream correction. Sub-problems (b) — potential geometry — and (c) — numerical scheme — remain open. They are the subject of Block V Phase A-0 / A-1 / A-2 / A-3 (arXiv v1 §5.3), which retains its priority.

A.8 Explicitly not claimed

- That MaoField solves any established open problem in theoretical physics, information retrieval, or AI.
- That the 50 localized patches per document correspond to specific ontological entities (“particles”, “atoms”, “concepts”). They are well-defined dynamical features of the PDE; their interpretation remains open.
- That the plateau at $dS/dt \approx 10$ is a rigorously defined thermodynamic NESS in the Seifert-Evans-Searles sense. It is a finite-time, finite-size, deterministic-integrator observation consistent with NESS phenomenology.
- That the kinematic self-similarity result (§A.3.3) is evidence of RG universality, a critical exponent, or membership in any known universality class. It is Kadanoff-block-averaging agreement between two stability-admissible time scales.
- That the architectural theorem (§A.4 (i)) implies BGE is a neutral upstream. It implies only that the base-source **mean** has no differential effect on the feedback dynamics; higher-order statistics were not ablated.
- That the Day 1–Day 4 compressed execution (2026-04-15, 36 hours) substitutes for a properly paced parameter-scan and robustness campaign. It does not.
- That the findings generalize beyond the specific `(, , N_hist, block_size) = (0.1, 0.05, 100, 2)` parameter choice used here. Parameter robustness is deferred to follow-up experiments (§A.9).

A.9 Follow-up

- **Parameter robustness:** `(, , N_hist)` scan across the ranges specified in the task specification `( {0.01, 0.1, 0.5}, {0, 0.05, 0.2}, N_hist {50, 100, 500})`. Do the qualitative findings hold?
- **Block-size dependence:** self-similarity exponent as a function of `block_size {2, 4, 8}`; distinguish seeding-hierarchy self-similarity from universal scaling.
- **True Langevin limit:** add noise $\cdot (x, t)$ and compare `c()`. The deterministic `c ≈ 10` is the $\rightarrow 0$ limit; the thermodynamic interpretation of the NESS in the stochastic case is sharper.
- **Retrieval correlation:** does the patch count correlate with document-level retrieval performance or semantic complexity? This is the natural test of the “multi-scale-seeded localized excitation = candidate semantic unit” interpretation.
- **True RG flow:** measure the feedback parameters `(_eff, _eff, D_eff)` at successive block scales and check for flow toward a fixed point. The

current experiment demonstrates kinematic self-similarity of configurations; a proper Wilson flow in the parameters is a separate observation.

- **History saturation:** run `t_sim = 100` to fully saturate `N_hist = 100` and confirm the qualitative findings.

A.10 Data and code availability

All analysis code (Rust engine source, Python analysis scripts), raw observable JSONs, stationary-state -field snapshots (`states.bin`, little-endian `f32` pairs), independent review records (code review, three observable-specific math reviews, exploration-signals analysis, verdict review), and the task specification originating this experiment are archived in `exp017_dialectics/results/phase_b_exp1/` and included in the v0.1.1 Zenodo release.

Linux Claude, 2026-04-15, under Yifan Chen's Gate 2 authorization. Independent review of this appendix is archived as `REVIEW_PHASE_B_EXP1_appendix.md`.